

INTERNSHIP PROJECT REPORT

Enterprise Networking Infrastructure

A Study on Network Topology Design, Inter-VLAN Routing & Security at Cisco
Systems

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Program: **B.Tech CSE (3CSE9)**

Institution: **IILM University**

Executive Summary & Overview

- **Industry Focus:** Internship centered on enterprise networking hardware, software, and telecommunications architecture at Cisco Systems.
- **Hands-on Training:** Practical engagement with Cisco IOS-based routers and Catalyst switches in simulated environment.
- **Core Competencies:** Focused on VLAN segmentation, inter-VLAN routing, dynamic routing protocols, and network access security.
- **Real-World Application:** Applied theoretical computer networking fundamentals to build scalable, fault-tolerant campus networks.



Organization Profile

Cisco Systems, Inc. — San Jose, California
Global Leader in Networking, Security, and Cloud
Telecommunications

Problem Statement & Objectives

Problem Statement

Modern enterprises face security risks, traffic congestion, and downtime when relying on unsegmented flat networks. The challenge was to design and simulate a resilient multi-department network that isolates departmental traffic while providing high availability and controlled inter-departmental routing.

Primary Objectives

Design a 3-layer hierarchical LAN topology using Cisco Packet Tracer, implement departmental VLAN isolation, configure Router-on-a-Stick / Layer 3 switching, enforce security policies via ACLs & Port Security, and evaluate dynamic routing protocol performance.

Hierarchical Architecture

Core Layer

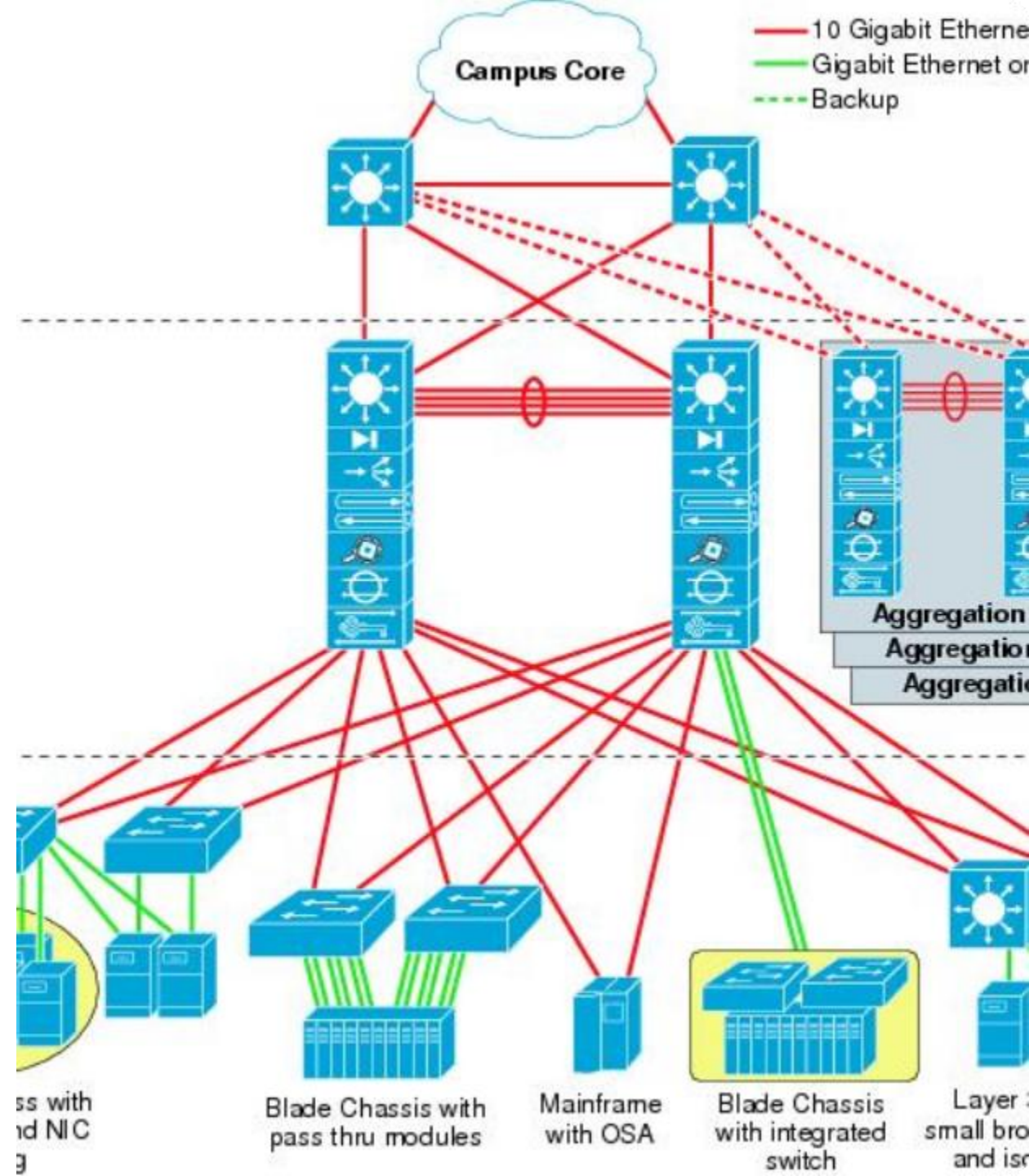
High-speed switching backbone ensuring low-latency packet forwarding across network blocks.

Distribution Layer

Multilayer switches performing Inter-VLAN routing, ACL policy enforcement, and packet filtering.

Access Layer

Layer 2 Catalyst switches connecting end devices directly with port-security enforcement.



VLAN & Subnet Addressing Scheme

VLAN ID	Department	IP Subnet (VLSM)	Subnet Mask	Default Gateway
VLAN 10	Administration	192.168.10.0 /26	255.255.255.192	192.168.10.1
VLAN 20	Human Resources	192.168.20.0 /26	255.255.255.192	192.168.20.1
VLAN 30	Engineering	192.168.30.0 /25	255.255.255.128	192.168.30.1
VLAN 40	Sales & Marketing	192.168.40.0 /26	255.255.255.192	192.168.40.1

Core Technologies & Protocols



L2 Switching

Configured IEEE 802.1Q trunking and Spanning Tree Protocol (STP) to eliminate switching loops while enabling multi-VLAN trunk links.



L3 Routing

Implemented Router-on-a-Stick subinterfaces and Switched Virtual Interfaces (SVIs) for seamless dynamic inter-VLAN routing.



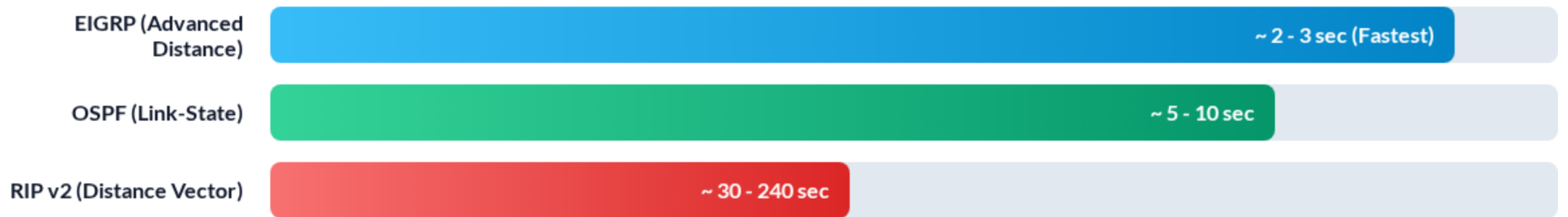
Network Security

Deployed MAC-based Port Security, SSH remote management, and standard/extended Access Control Lists (ACLs) to filter traffic.

Project Implementation Methodology



Routing Protocol Convergence Comparison



Evaluation of failover and convergence times demonstrated EIGRP and OSPF's superiority over RIP v2 for scalable enterprise deployments.

Security & Policy Enforcement



Port Security: Restricted access layer switch ports using sticky MAC address binding to prevent unauthorized rogue device access.



Access Control Lists (ACLs): Applied extended IP ACLs on router subinterfaces to prevent unauthorized traffic between Sales and HR VLANs.



Encrypted Administration: Disabled unencrypted Telnet and configured SSH v2 with RSA key pairs for secure CLI administration.



Native VLAN Security: Changed default VLAN 1 to dedicated unused VLAN ID to mitigate VLAN hopping attacks.

Key Technical Outcomes

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Isolated Departmental VLANs

100%

Inter-VLAN Reachability

Project Achievements

The simulated infrastructure achieved seamless packet flow across subnets while strict security rules prevented unauthorized inter-departmental communications.

Demonstrated practical mastery in configuring Cisco IOS devices, diagnosing physical/data link layer issues, and designing extensible corporate networks.

Questions & Discussion

Thank You!

Yash Pandey | B.Tech Computer Science & Engineering

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Project Supervised under Cisco Systems Internship Program

Image Sources



<https://cdn.shopify.com/s/files/1/0736/7566/9719/files/enterprise-ai-infrastructure-data-center-wifi7-local-ai-2026.jpg>

Source: network-switch.com



Thumbnail

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